Assignment 7

Total = 10 Marks

Multiple Choice Questions

Question 1: Principal Component Analysis (PCA) transforms original variables into a new set of variables called principal components (PC's). These PC's are:

(a) correlated and non-orthogonal
(b) orthogonal and correlated
(c) orthogonal and uncorrelated
(d) None of these

[1 *Mark*]

Correct answer: (c) orthogonal and uncorrelated

Detailed Solution:

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms a set of correlated variables into a new set of variables called principal components. These principal components are linear combinations of the original variables.

The principal components have the following properties:

They are orthogonal (perpendicular) to one another, meaning that each principal component represents a unique direction of variation in the data.
They are uncorrelated, meaning that the covariance between any pair of principal components is zero.

Question 2: In Principal Component Analysis (PCA), the principal component corresponding to the largest eigenvalue represents the:

(a) direction of minimum variance in the dataset
(b) direction of maximum variance in the dataset
(c) direction of average variance in the dataset
(d) None of these

[1 *Mark*]

Correct answer: (b) direction of maximum variance in the dataset

Detailed Solution:

In Principal Component Analysis (PCA), the eigen vectors are calculated from the covariance (or correlation) matrix of the data. Each eigenvector represents the direction of a principal component. The corresponding eigenvalue measures the amount of variance explained by that principal component. The largest eigenvalue corresponds to the first principal component (PC1), which explains the maximum variance in the dataset. Subsequent principal components explain progressively smaller amounts of the remaining variance while remaining orthogonal to the preceding components.

Question 3: In Principal Component Analysis (PCA), the principal components are obtained from the:

(a) mean vector of the dataset
(b) correlation between dependent variables
(c) covariance (or correlation) matrix of the data
(d) None of these

[1 *Mark*]

Correct answer: (c) covariance (or correlation) matrix of the data

Detailed Solution:

In Principal Component Analysis (PCA), the first step is to compute either the covariance matrix or the correlation matrix of the data. An eigenvalue decomposition of this matrix is then performed. The eigenvectors of the covariance (or correlation) matrix define the principal components. The eigenvalues indicate the amount of variance explained by each principal component.

Question 4: The sum of all eigenvalues of a covariance matrix is equal to:

(a) trace of the covariance matrix
(b) determinant of the covariance matrix
(c) product of diagonal elements of the covariance matrix
(d) None of these

[1 *Mark*]

Correct answer: (a) trace of the covariance matrix.

Detailed Solution:

For any square matrix, the sum of its eigenvalues is equal to its trace, which is the sum of its diagonal elements. For a covariance matrix, the diagonal elements represent the variances of the individual variables. Therefore, the trace of the covariance matrix equals the total variance of the dataset.

Question 5: Using the cumulative explained variance criterion, the number of PCs (k) is selected such that:

(a) eigenvalue of the kth PC is maximum
(b) cumulative explained variance $\geq$ threshold value
(c) both (a) and (b)
(d) None of these

[1 *Mark*]

Correct answer: (b) cumulative explained variance $\geq$ threshold value

Detailed Solution:

In Principal Component Analysis (PCA), the principal components are ordered according to the amount of variance they explain. A common approach for selecting the number of principal components is the cumulative explained variance criterion.

The smallest number of principal components, k, is chosen such that,

$$\frac{\sum_{i=1}^{k} \lambda_i}{\sum_{i=1}^{p} \lambda_i} \geq \tau$$

where, λ_i is the ith eigenvalue, p is the total number of principal components and τ is the threshold chosen.

This criterion ensures that the selected principal components retain most of the variability present in the original dataset while reducing its dimensionality.

Question 6: In PCA, a scree plot is a plot between:

(a) principal components and percentage of total variance explained
(b) observations and principal components
(c) principal components and sample means
(d) covariance matrix and correlation matrix

[1 *Mark*]

Correct answer: (a) principal components and or percentage of total variance explained

Detailed Solution:

A scree plot is a graphical tool used in Principal Component Analysis (PCA) to determine the appropriate number of principal components to retain.

The plot displays:

X-axis: Principal components (PC_1, PC_2, PC_3, ...)

Y-axis: Corresponding percentage of total variance explained by each principal component.

Question 7: In Empirical Orthogonal Function (EOF) analysis, the original geophysical dataset is typically arranged in a:

(a) One-dimensional matrix
(b) Three-dimensional array consisting of two spatial dimensions and one temporal dimension
(c) Two-dimensional matrix
(d) One-dimensional mean vector of all variables

[1 *Mark*]

Correct answer: (b) Three-dimensional array consisting of two spatial dimensions and one temporal dimension

Detailed Solution:

Empirical Orthogonal Function (EOF) analysis is widely used to identify dominant patterns of variability in spatiotemporal datasets. The original dataset is typically stored as a three-dimensional array, consisting of: Latitude, Longitude, and Time. Thus, each observation corresponds to a spatial field observed at a particular time. Before performing EOF analysis, the three-dimensional dataset is reshaped into a two-dimensional matrix, with one dimension representing time and the other representing all spatial grid points.

Question 8: In a two-way ANOVA, Attribute 1 has $a = 5$ and $SSA = 120$, while Attribute 2 has $b = 4$ and $SSB = 80$. The sum of squares due to error is $SSE = 180$. Calculate the F-statistic for Attribute-1.

(a) 2
(b) 5
(c) 6
(d) 8

[3 *Mark*]

Correct answer: (a) 2

Detailed Solution:
In a two-way ANOVA, the F-statistic for Attribute-1 is computed as

$$F_1 = \frac{MSA_1}{MSE}$$

where,

$$MSA_1 = \frac{SSA_1}{a-1} = \frac{120}{5-1} = \frac{120}{4} = 30$$

$$MSE = \frac{SSE}{(a-1)(b-1)} = \frac{180}{4 \times 3} = 15$$

$$F_1 = \frac{MSA_1}{MSE} = \frac{30}{15} = 2$$

Therefore, the F-statistic for Attribute-1 is 2

Therefore, the correct answer is: 2.